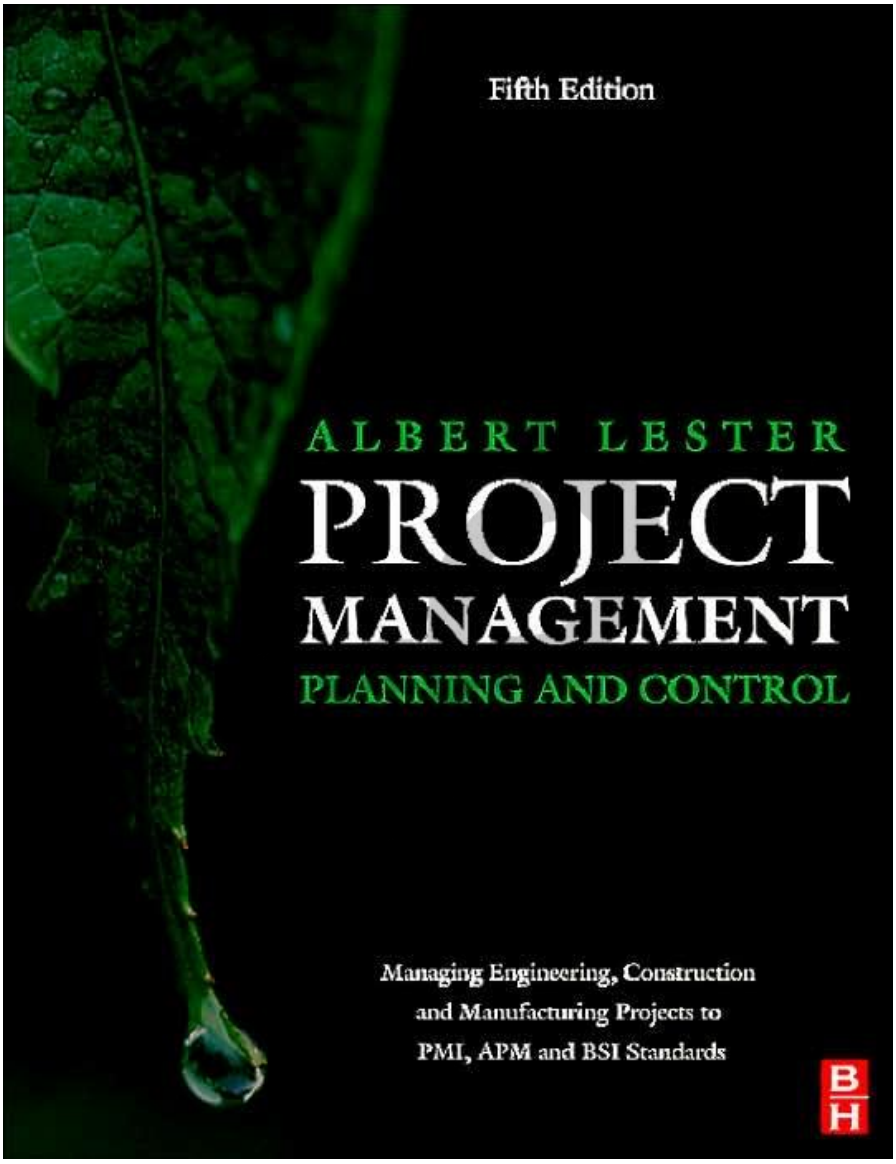
Project Management, Planning and Control:

Managing Engineering, Construction and Manufacturing Projects to PMI, APM and BSI Standards *by* *Albert Lester*



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Cross reference to APM and PMI bodies of knowledge

Chapter No.	Topic	APM. BoK No.	PMI BoK No.
1	Project definition	1.1,	1.2
2	Project management	1.1, 1.6	1.3
3	Programme management	1.2, 1.3	
4	Project context (project environment)	1.4	1.6
5	Business case	5.1, 1.5, 4.1	
6	Investment appraisal	5.1	
7	Stakeholder management	2.2	2.2, 10.4
8	Project success criteria	2.1	
9	Organization structures	6.7	2.3
10	Organizational roles	6.8	2.3
11	Project life cycles	6.1	2.1
12	Work breakdown structures	3.1	5.3
13	Planning blocks and subdivision of blocks	3.2	
14	Estimating	4.3	7.1
15	Project management plan	2.4, 6.9	4.3
16	Risk management	2.5	11.1
17	Quality management	2.6	8.1
18	Change management	3.5, 3.8	4.3, 5.5
19	Configuration management	4.7	5.4
20	Basic network principles	3.2	6.2
21	Analysis types	3.2	6.3
22	Precedence or activity on node (AoN) diagrams	3.2	6.4
23	Lester diagram	3.2	
24	Float	3.2	6.3
25	Milestones and line of balance	3.2	6.4
26	Simple examples	3.2	
27	Progress reporting	3.2	10.3
28	Project management and planning	3.2	6.2
29	Network applications outside the construction industry	3.2	

30	Resource loading	3.3,	7.1
31	Cash flow forecasting	3.4	7.3
32	Cost control and EVA	3.6	7.3, 10.3
33	Control graphs and reports	3.6	7.4, 10.3
34	Procurement	5.4	12.1
35	Value management	2.3, 4.5	(5.2)
36	Health and safety and environment	2.7	
37	Information management	3.7	10.2
38	Communication	7.1	10.1
39	Team building	7.2	9.3
40	Leadership	7.3, 7.7, 7.9	(2.4)
41	Negotiation	7.5	(2.4)
42	Conflict management and dispute resolution	7.4	
43	Project close-out and hand over	6.5	12.6
44	Project close-out report and review	6.6	10.3

( ) = Discussed in context of other topic.

Foreword to the first edition

by Geoffrey Trimble, Professor of Construction Management,
University of Technology, Loughborough

A key word in the title of this book is 'control'. This word, in the context of management, implies the observation of performance in relation to plan and the swift taking of corrective action when the performance is inadequate. In contrast to many other publications which purport to deal with the subject, the mechanism of control permeates the procedures that Mr Lester advocates. In some chapters, such as that on Manual and Computer Analysis, it is there by implication. In others, such as that on Cost Control, it is there in specific terms.

The book, in short, deals with real problems and their real solutions. I commend it therefore both to students who seek to understand the subject and to managers who wish to sharpen their performance.

Preface

The shortest distance between two points is a straight line
Euclid

The longest distance between two points is a shortcut
Lester

The first edition of this book dealt mainly with the fundamentals and industrial applications of network analysis and a cost/progress control technique called SMAC, which is now known universally as earned value analysis.

In the light of the rapid advances in computers, especially the development of the PC, the second edition updated these techniques and included a detailed description of a well-known computerized project management program.

The third edition expanded the earned value section, described two other computer project management programs and introduced some of the other 'hard topics' required by a project manager.

Because of the demand created by students taking the Association for Project Management's APMP examination, the fourth edition included all the hard topics required in the examination syllabus. The soft topics were deliberately not included, as these were applicable to general management and not exclusive to project management. To illustrate how these hard topic techniques can be applied and incorporated in practice, several fully worked examples of typical projects were included.

As with the previous volumes, this fifth edition was written to meet a specific need. In this case it was the fulfilment of a request by the publishers and some of my lecturing colleagues at University College London, to produce a book which included all the hard and soft topics required by the latest syllabus of the APMP examination. In addition the book should also meet the needs of the PMI examination as stipulated in the PMI Body of Knowledge.

When starting a new edition, one inevitably wonders whether any section of the current edition has become obsolete and whether it should therefore be updated or left out altogether. After all, a plethora of computer-generated coloured printouts such as Gantt charts showing base schedules and updates, tables, summaries, histograms, pie charts, 'S' curves and even networks themselves have replaced hand-drawn or typed documents. My first thoughts were therefore to leave out the chapters on arithmetical analysis and the case for manual analysis since nearly all network planning is now carried out by sophisticated computer programs which not only take the chores out of the analysis process, but also enable 'what if' scenarios to be rapidly examined.

However, we have not yet reached the stage when computers can think for themselves, so that the creation of the logic of a network must still be done by humans. Except for standard repetitive projects where it is possible to design logic modules, each project network of any